A spectral approach to arithmetic correlations for binary FCSR sequences with prime connection integers

Feifei Yan[1], Pinhui Ke[2*], Chenhuang Wu[3]

[1]School of Mathematics and Statistics, Fujian Normal University, Fuzhou, 350117, Fujian, China.
[2*]Key Laboratory of Analytical Mathematics and Applications (Ministry of Education), Fujian Normal University, Fuzhou, 350117, Fujian, China.
[3]Provincial Key Laboratory of Applied Mathematics, Putian University, Putian, 351100, Fujian, China.

*Corresponding author(s). E-mail(s): keph@fjnu.edu.cn;
Contributing authors: ffyfjnu@139.com; ptuwch@163.com;

Abstract

Arithmetic correlation is an important metric for measuring feedback with carry shift register sequences, and its value should be as small as possible. For binary FCSR sequences with a prime connection integer $\boldsymbol{p}$ and for which $\mathbf{ord}_{\boldsymbol{p}}\mathbf{(2)}$ is odd, where $\mathbf{ord}_{\boldsymbol{p}}\mathbf{(2)}$ is the order of $\mathbf{2}$ modulo $\boldsymbol{p}$, the arithmetic correlation can take on any value from the set of differences obtained by subtracting, for each set in the collection consisting of the subgroup generated by $\mathbf{2}$ and all its cosets, the number of odd representatives from the number of even representatives within that set. From this perspective, we develop a unified spectral method for arithmetic correlation, derive an upper bound on it, and establish conditions under which it takes small values. We also analyze cases with a prime connection integer $\boldsymbol{p}$ where the number of cosets is $\mathbf{2}$, $\mathbf{4}$, and $\mathbf{6}$, respectively, and characterize when the arithmetic correlation takes small values.

Keywords: FCSR sequences, Arithmetic correlation, Connection integer, Fourier analysis

1 Introduction

Peudorandom sequences generated by feedback shift registers are widely used in communications and stream cipher analysis [1–3]. In the 1990s, Goresky, Klapper and Xu introduced the feedback with carry shift register (FCSR)[3–5]. Compared with linear feedback shift registers (LFSRs), the FCSR is an analog of the LFSR with the addition of a carry operation. It incorporates a 2-adic number theoretic structure while maintaining the advantages of simple hardware implementation and relatively high speed. As a result, it has quickly become an important subject of research in sequence theory and cryptography[3].

FCSR sequences have two important measures: N-adic complexity and arithmetic correlation. The concept of arithmetic autocorrelation was first introduced by Mandelbaum[6] in the study of arithmetic codes and was later extended to FCSR sequences by Goresky and Klapper[3, 7].

Let $\mathcal{S} = (s_i)_{i\geq 0}$ be a T-periodic binary sequence defined over $\mathbb{F}_2$, generated by an FCSR. According to the FCSR theory [3], $\mathcal{S}$ corresponds uniquely to a 2-adic number

$$\alpha = \sum_{i=0}^{\infty} s_i 2^i = -\frac{\sum_{i=0}^{T-1} s_i 2^i}{2^T - 1} = -\frac{f}{q},$$

where $0 < f < q$ and $\gcd(f, q) = 1$. The integer q is called the *connection integer* of the sequence $\mathcal{S}$ and $q = \frac{2^T-1}{\gcd(\sum_{i=0}^{T-1} s_i 2^i, 2^T-1)}$. Equivalently, $\mathcal{S}$ can also be given by [3]

$$s_i = \left(f \cdot 2^{-i} \bmod q\right) (\mathrm{mod} 2). \tag{1}$$

The period of $\mathcal{S}$ is $T = \mathrm{ord}_q(2)$, denote the multiplicative order of 2 modulo q. Now let $\mathcal{T} = (t_i)_{i\geq 0}$ be another binary sequence with the same period T and the same connection integer q. For any shift τ, denote the τ-shift of $\mathcal{T}$ by $\mathcal{T}^{(\tau)}$. And $\mathcal{T}^{(\tau)}$ corresponds to the 2-adic number

$$\beta^{(\tau)} = \sum_{i=0}^{\infty} t_{i+\tau} 2^i = -\frac{\sum_{i=0}^{T-1} t_{i+\tau} 2^i}{2^T - 1} = -\frac{h_\tau}{q},$$

where $\gcd(h_\tau, q) = 1$. From α and $\beta^{(\tau)}$ we define a new binary sequence $\mathcal{U} = (u_i)_{i\geq 0}$ satisfying

$$\alpha - \beta^{(\tau)} = \sum_{i=0}^{\infty} (s_i - t_{i+\tau}) 2^i = \sum_{i=0}^{\infty} u_i 2^i.$$

The *arithmetic crosscorrelation* of $\mathcal{S}$ and $\mathcal{T}$ at shift τ, denoted $C^A_{\mathcal{S},\mathcal{T}}(\tau)$, is defined as the number of zeros minus the number of ones in one full period of $\mathcal{U}$ with period T [3]. In particular, when $\mathcal{S} = \mathcal{T}$, $\mathcal{C}^A_{\mathcal{S},\mathcal{S}}(\tau)$ is called the *arithmetic autocorrelation* of $\mathcal{S}$ at τ, denoted $\mathcal{A}^A_{\mathcal{S}}(\tau)$.

Recently, arithmetic autocorrelation of FCSR sequences has attracted renewed attention. Current research includes studies on arithmetic correlation for m-sequences [8–11] and half-ℓ-sequences [12–14], as well as on ideal arithmetic correlation [15, 16].

In particular, the arithmetic correlation properties of binary FCSR sequences are closely related to the multiplicative subgroup generated by 2 modulo the connection integer [3]. Based on this, Chen et al. [15] derived some sufficient conditions for ideal arithmetic correlation. Later, Edemskiy [16] proved these conditions are also necessary. However, their work is limited to the case where the order of 2 modulo the connection integer is even. The case where the order of 2 modulo the connection integer is odd remains an open problem. And arithmetic correlation values should be as small as possible, which leads us to further examine cases with small arithmetic correlation, such as when its absolute value is 1, or 1 and 3.

Inspired by the aforementioned study, this paper focuses on binary FCSR sequences whose connection integer is an odd prime p. Connecting with the ideal arithmetic autocorrelation problem discussed in [16], we systematically reformulate the arithmetic autocorrelation using a coset parity difference framework. From a broader perspective, this paper attempts to unify the seemingly scattered integer phenomena in FCSR arithmetic autocorrelation into a coherent explanation based on coset parity structure and odd character Fourier decomposition on quotient groups. This viewpoint is not only compatible with the existing analysis of ideal cases in [16], but also provides a unified framework for the related phenomena, which is more conductive to structural investigation. Based on this framework, we first derive an upper bound on the arithmetic correlation and establish conditions for the small arithmetic correlation values. Then, as specific applications, we provide a detailed analysis for the cases of 2, 4, and 6 cosets, respectively.

The paper is organized as follows. Section 2 establishes the coset parity interpretation for prime connection integers. When the order of 2 modulo odd prime connection integer p is odd, Section 3 derives the properties of coset parity differences, presents their Fourier expansion for the prime case, and establishes both an upper bound and conditions for achieving small arithmetic correlation. Section 4 analyzes the cases where the number of cosets is 2, 4 or 6. Section 5 concludes the paper.

2 Preliminaries

This paper focuses on the binary case and restricts the connection integer to an odd prime p. In this situation, every nonzero residue modulo p is a unit, which allows the problem to be treated uniformly within a single multiplicative group and its coset structure. It is precisely this uniformity that makes the prime case tidier than that of a general odd connection integer.

For binary FCSR sequence $\mathcal{S}$ generated by the 2-adic number $\alpha = -\frac{f}{p}$, where $1 \leq f \leq p-1$. Denote

$$G = \mathbb{F}_p^*, \qquad D = \langle 2 \rangle \subseteq G, \qquad d = \mathrm{ord}_p(2) = |D|, \qquad m = \frac{p-1}{d},$$

where $\mathbb{F}_p^*$ denotes the multiplicative group of the finite field $\mathbb{F}_p = \{0, 1, \ldots, p-1\}$, and $\langle 2 \rangle$ represents the multiplicative subgroup generated by 2. From Eq. (1) we have

$$s_i = \left(f \cdot 2^{-i} \bmod p\right) (\mathrm{mod} 2).$$

Hence, as i runs through a period of $\mathrm{ord}_p(2)$, the elements $f \cdot 2^{-i} \bmod p$ traverse exactly each element of the coset fD once. Consequently, the difference between the number of zeros and the number of ones in one period is equal to the difference between the number of standard representatives in the coset fD that are even and the number that are odd. Here, the standard representative of an element in $\mathbb{F}_p^*$ is taken as the least nonnegative residue modulo p.

To perform the subsequent Fourier analysis, fix a multiplicative character χ of order m with kernel D. Fix a primitive mth root of unity ζ_m and all cosets of D by

$$C_j = \{x \in G : \chi(x) = \zeta_m^j\} \qquad j = 0, 1, \dots, m-1.$$

Definition 1 For $j = 0, 1, \dots, m-1$, C_j be the multiplicative coset of D in G. Define the *coset parity difference* of C_j as

$$F(C_j) = \big|\{x \in C_j : \text{standard representative of } x \text{ is even}\}\big| - \big|\{x \in C_j : \text{standard representative of } x \text{ is odd}\}\big|.$$

And we write $F_j = F(C_j)$ for convenience. For each $x \in G$, define

$$\varepsilon(x) = \begin{cases} 1, & \text{standard representative of } x \text{ is even}, \\ -1, & \text{standard representative of } x \text{ is odd}. \end{cases}$$

Then

$$F_j = \sum_{x \in C_j} \varepsilon(x).$$

Furthermore, for $1 \le r \le m-1$, define the half interval character sum

$$H_r = \sum_{n=1}^{(p-1)/2} \chi^r(n). \tag{2}$$

The following proposition gives the exact correspondence between arithmetic correlation and coset parity difference in the prime connection integer case. Here we reorganize it into a form suitable for subsequent Fourier analysis on the coset structure.

Proposition 1 *Let $\mathcal{S}$ and $\mathcal{T}$ be two binary FCSR sequences with the same connection integer, which is an odd prime p, corresponding to the 2-adic numbers $\alpha = -\frac{f}{p}$ and $\beta = -\frac{g}{p}$, respectively. For any $1 \le \tau \le \mathrm{ord}_p(2) - 1$, the τ-shift of $\mathcal{T}$ corresponds to $\beta^{(\tau)} = -\frac{h_\tau}{p}$, with*

$$h_\tau \equiv g \cdot 2^{-\tau} \pmod{p}.$$

Let $a_\tau \in \{1, \dots, p-1\}$ be the residue determined by

$$a_\tau \equiv f - h_\tau \equiv f - g \cdot 2^{-\tau} \pmod{p},$$

then

$$\mathcal{C}^A_{\mathcal{S},\mathcal{T}}(\tau) = F(a_\tau D).$$

Proposition 1 gives $a_\tau D$ equals one of the cosets C_j, and consequently $\mathcal{C}^A_{\mathcal{S},\mathcal{T}}(\tau) = F_j$ for some j. Thus, in the case of an odd prime connection integer, the arithmetic correlation at a nontrivial shift can be rewritten as a parity difference on the corresponding coset. It expresses the correlation as a function defined on the cosets, whose spectral expansion via Fourier analysis on the set of cosets will be the basis for our subsequent analysis.

3 The structure of arithmetic correlation values in the odd order case

If $\mathrm{ord}_p(2)$ is even, then $-1 \in D$. This implies that the number of even representatives equals the number of odd representatives in each coset, and thus the parity difference of every coset C_j is zero. That is, $F_j = 0$ for all j. Hence, every binary FCSR sequence with connection integer p has ideal arithmetic correlation. More precisely, $\mathcal{C}^A_{\mathcal{S},\mathcal{T}}(\tau) = 0$ when $\mathcal{S} \neq \mathcal{T}^{(\tau)}$ for all $\tau \geq 0$. Detailed discussions of this case can be found in [15, 16]. From now on we assume $d = \mathrm{ord}_p(2)$ is odd.

3.1 Distribution parity differences in cosets

Lemma 1 Assume $d = \mathrm{ord}_p(2)$ is odd. Then for every j,
(i) $F_j \equiv d \pmod 2$. That is, F_j is odd.
(ii)$|F_j| \leq d$, and $F_j \in \{-d, -(d-2), \ldots, -1, 1, \ldots, d-2, d\}$.
(iii) $\sum_{j=0}^{m-1} F_j = 0$.

Proof (i) Write e_j and o_j for the numbers of even and odd standard representatives in C_j, respectively. Then $e_j + o_j = |C_j| = d$ and $F_j = e_j - o_j$. When computing modulo 2,

$$F_j = e_j - o_j \equiv e_j + o_j \equiv d \pmod 2.$$

Since d is odd, F_j is odd.

(ii) The inequality $|F_j| = |e_j - o_j| \leq e_j + o_j = d$ is immediate. Combined with (i), we obtain the stated set of possible values.

(iii) The cosets $C_0, \ldots, C_{m-1}$ partition G. Hence

$$\sum_{j=0}^{m-1} F_j = |\{1 \leq x \leq p-1 : x \text{ even}\}| - |\{1 \leq x \leq p-1 : x \text{ odd}\}|.$$

Among $1, 2, \ldots, p-1$ there are exactly $\frac{p-1}{2}$ even numbers and $\frac{p-1}{2}$ odd numbers, so the sum is zero. □

Lemma 1 has significant consequences for the arithmetic correlation of sequence. Since each nontrivial shift value equals some F_j, we immediately obtain:

Corollary 1 Let $d = \mathrm{ord}_p(2)$ be odd. Let $\mathcal{S}$ and $\mathcal{T}$ be two binary FCSR sequences with the same connection integer p. Then the arithmetic correlation between $\mathcal{S}$ and $\mathcal{T}$ satisfies:
(i) All nontrivial shift arithmetic correlation values are odd.

(ii) Each such value lies in the set $\{-d, -(d-2), \ldots, -1, 1, \ldots, d-2, d\}$.
(iii) The sum of all nontrivial shift arithmetic correlation values is zero.

It shows that when d is odd, 0 cannot appear as a nontrivial shift arithmetic correlation value. This is the exact opposite of the ideal autocorrelation that occurs when d is even.

Lemma 2 If $d = \mathrm{ord}_p(2)$ is odd, then $m = \frac{p-1}{d}$ is even, and $-1 \notin D$.

Proof Since $p-1$ is even and d is odd, m is even. If $-1 \in D$, then D contains an element of order 2, forcing its size d to be even, a contradiction. Hence $-1 \notin D$. □

Lemma 3 If $d = \mathrm{ord}_p(2)$ is odd, then for every j we have

$$F_{j+m/2} = -F_j,$$

where the subscript is taken modulo m.

Proof By Lemma 2, m is even and $-1 \notin D$. Because $\ker \chi = D$, we have $\chi(-1) \neq 1$, and $\chi(-1)^2 = \chi(1) = 1$, so $\chi(-1) = -1$, that is $\chi(-1) = \zeta_m^{m/2}$. Hence multiplication by -1 sends C_j to $C_{j+m/2}$.

Now consider the map $x \mapsto p - x$, which modulo p is exactly $x \mapsto -x$. It therefore gives a bijection from C_j to $C_{j+m/2}$. Since p is odd, x and $p-x$ have opposite parity, so ε changes sign under this map. Consequently,

$$F_{j+m/2} = \sum_{x \in C_{j+m/2}} \varepsilon(x) = \sum_{x \in C_j} \varepsilon(p-x) = -\sum_{x \in C_j} \varepsilon(x) = -F_j.$$

□

Lemma 3 shows that in the odd order case, the coset parity difference vector is always determined by its first half, the second half being just the negatives of the first half. This antisymmetry has powerful consequences for the distribution of small values.

Corollary 2 Assume $d = \mathrm{ord}_p(2)$ is odd. For any positive odd integer a, the number of indices j with $F_j = a$ equals the number of indices with $F_j = -a$.

Proof The map $j \mapsto j+m/2$ is an involution on the index set $\{0, 1, \ldots, m-1\}$, and by Lemma 3 it sends a value a to $-a$ (and vice versa). Hence the two values occur equally often. □

Thus, if a small positive value appears among the F_j, its negative must appear exactly as many times. We now foucues on what happens when all parity differences are bounded by a small constant. The simplest nontrivial bounds are $|F_j| \leq 1$ and $|F_j| \leq 3$.

Corollary 3 Assume $d = \mathrm{ord}_p(2)$ is odd. If $|F_j| \le 1$ for all j, then necessarily

$$F_j \in \{1, -1\} \quad \text{for every } j,$$

and the numbers of 1 and -1 are both exactly $\frac{m}{2}$.

Proof By Lemma 1 (i), each F_j is odd. The only odd integers with absolute value at most 1 are ± 1. Hence $F_j \in \{1, -1\}$. Corollary 2 forces the counts of 1 and -1 to be equal, since there are m indices total, each count must be $\frac{m}{2}$. □

Corollary 4 Assume $d = \mathrm{ord}_p(2)$ is odd. If $|F_j| \le 3$ for all j, then

$$F_j \in \{1, -1, 3, -3\} \quad \text{for every } j,$$

and the pairs $(1, -1)$ and $(3, -3)$ occur equally often.

Proof By Lemma 1 (i), F_j is odd, and the only odd integers with $|F_j| \le 3$ are $\pm 1, \pm 3$. Corollary 2 guarantees that the positive and negative values of each absolute value appear the same number of times. □

These two corollaries show that the first two levels of small parity differences are extremely rigid, they can only be ± 1 or $\pm 1, \pm 3$, and in both cases the values must appear in symmetric pairs.

Corollary 5 Let $d = \mathrm{ord}_p(2)$ be odd. Let $\mathcal{S}$ and $\mathcal{T}$ be two binary FCSR sequences with connection integer p.
(i) If for every j has $|F_j| \le 1$, then all nontrivial shift arithmetic correlation values between $\mathcal{S}$ and $\mathcal{T}$ are ± 1.
(ii) If for every j has $|F_j| \le 3$, then all nontrivial shift arithmetic correlation values between $\mathcal{S}$ and $\mathcal{T}$ belong to $\{\pm 1, \pm 3\}$.

3.2 Fourier expansion of coset parity differences with applications to arithmetic correlation

Lemma 4 With the notation above. If $d = \mathrm{ord}_p(2)$ is odd, then $\chi(-1) = -1$. For every even r with $1 \le r \le m - 1$, we have $H_r = 0$. Hence the Fourier expansion is

$$F_j = \frac{2}{m} \sum_{\substack{1 \le r \le m-1 \\ r \text{ odd}}} H_r \, \zeta_m^{-jr}.$$

Proof Denote $\mathbf{1}_j$ be the indicator function of the coset C_j. By the orthogonality of characters of a finite cyclic group [17, Chapter 5], for any $x \in G$,

$$\mathbf{1}_j(x) = \frac{1}{m} \sum_{r=0}^{m-1} \zeta_m^{-jr} \chi^r(x).$$

On the other hand, the even standard representatives are $2, 4, \dots, p-1$ and the odd ones are $1, 3, \dots, p-2$. Hence

$$F_j = \sum_{n=1}^{(p-1)/2} \big[\mathbf{1}_j(2n) - \mathbf{1}_j(-2n)\big].$$

Since $2 \in D = \ker\chi$, we have $\chi(2) = 1$, so multiplication by 2 does not change the coset, therefore

$$\mathbf{1}_j(2n) = \mathbf{1}_j(n), \qquad \mathbf{1}_j(-2n) = \mathbf{1}_j(-n).$$

Consequently,

$$F_j = \sum_{n=1}^{(p-1)/2} \big[\mathbf{1}_j(n) - \mathbf{1}_j(-n)\big].$$

Substituting the Fourier expansion of the indicator function yields

$$F_j = \frac{1}{m} \sum_{r=0}^{m-1} \zeta_m^{-jr} \sum_{n=1}^{(p-1)/2} \big[\chi^r(n) - \chi^r(-n)\big].$$

Because $\chi^r(-n) = \chi^r(-1)\chi^r(n)$, this becomes

$$F_j = \frac{1}{m} \sum_{r=0}^{m-1} \big(1 - \chi^r(-1)\big) H_r\, \zeta_m^{-jr}.$$

When $r = 0$, the coefficient $1 - \chi^0(-1) = 0$, so that term vanishes. By Lemma 2, $-1 \notin D = \ker\chi$, so $\chi(-1) \neq 1$. Because $(-1)^2 = 1$, we must have $\chi(-1) = -1$. Now take an even r. Then $\chi^r(-1) = (-1)^r = 1$, thus χ^r is an even character. Since $1 \le r \le m-1$, χ^r is nontrivial, so its sum over G is zero [17, Theorem 5.4]. Pairing n with $-n$ gives

$$0 = \sum_{x \in G} \chi^r(x) = 2 \sum_{n=1}^{(p-1)/2} \chi^r(n) = 2H_r,$$

hence $H_r = 0$. Then $1 - \chi^r(-1) = 1 - (-1)^r$, which is 0 for even r and 2 for odd r. $\square$

From Lemma 4 it follows that for every odd integer $r\,(1 \le r \le m-1)$, we have the Fourier inversion

$$H_r = \frac{1}{2} \sum_{j=0}^{m-1} F_j \zeta_m^{jr}. \tag{3}$$

Theorem 2 *Assume $d = \mathrm{ord}_p(2)$ is odd. Then*

$$\sum_{j=0}^{m-1} F_j^2 = \frac{4}{m} \sum_{\substack{1 \le r \le m-1 \\ r \text{ odd}}} |H_r|^2.$$

Proof According to Lemma 4, define

$$\widehat{F}(r) = \begin{cases} 2H_r, & r \text{ odd}, \\ 0, & r \text{ even or } r = 0. \end{cases}$$

Then the Fourier expansion can be written as $F_j = \frac{1}{m}\sum_{r=0}^{m-1} \widehat{F}(r)\zeta_m^{-jr}$. By the Parseval-Plancherel formulas [18], we have

$$\sum_{j=0}^{m-1} |F_j|^2 = \frac{1}{m}\sum_{r=0}^{m-1} |\widehat{F}(r)|^2.$$

Since the F_j are real integers, $|F_j|^2 = F_j^2$. The right hand side receives contributions only from odd r, whence

$$\sum_{j=0}^{m-1} F_j^2 = \frac{1}{m}\sum_{\substack{1\le r\le m-1\\ r \text{ odd}}} |2H_r|^2 = \frac{4}{m}\sum_{\substack{1\le r\le m-1\\ r \text{ odd}}} |H_r|^2.$$

□

From Theorem 2, an arithmetic correlation upper bound can be derived for the case where the connection integer is an odd prime p and $\mathrm{ord}_p(2)$ is odd.

Theorem 3 *Let $\mathcal{S} = (s_i)_{i\ge 0}$ and $\mathcal{T} = (t_i)_{i\ge 0}$ be two binary periodic FCSR sequences with the same odd prime connection integer p and $\mathrm{ord}_p(2)$ is odd. Then for any nontrivial arithmetic correlation, we have*

$$\left|\mathcal{C}^A_{\mathcal{S},\mathcal{T}}(\tau)\right| \ll \sqrt{p}\,\log p,$$

where $A \ll B$ denotes $|A| \le cB$ for an absolute constant $c > 0$.

Proof By Lemma 4, $\chi(-1) = -1$, so when r is odd, χ^r is an odd character. For prime modulus, χ^r is an odd primitive character. According to [19], for each odd r with $1 \le r \le m-1$ we have

$$H_r = -\frac{i}{\pi}\, G\big(\chi^r\big)\, L\big(1, \overline{\chi}^r\big), \tag{4}$$

where $i = \sqrt{-1}$,

$$G\big(\chi^r\big) = \sum_{t=1}^{p-1} \chi^r(t) e^{2\pi i t/p}$$

is the Gauss sum, and

$$L\big(1, \overline{\chi}^r\big) = \sum_{n=1}^{\infty} \overline{\chi}^r(n) n^{-1}$$

is the Dirichlet L-function. For a prime modulus p, the Gauss sum of a nontrivial character satisfies $|G(\chi^r)| = \sqrt{p}$ [20, Proposition 8.2.2.]. Hence

$$|H_r|^2 = \frac{p}{\pi^2}\left|L\big(1, \overline{\chi}^r\big)\right|^2, \qquad 1 \le r \le m-1,\ r \text{ odd}.$$

By Theorem 2, substituting the expression for $|H_r|^2$ yields

$$\frac{p}{\pi^2}\sum_{\substack{1\le r\le m-1\\ r \text{ odd}}} \left|L\big(1, \overline{\chi}^r\big)\right|^2 = \frac{m}{4}\sum_{j=0}^{m-1} F_j^2. \tag{5}$$

From Proposition 1, we have

$$\left|\mathcal{C}^A_{\mathcal{S},\mathcal{T}}(\tau)\right| \le \max_{0\le j\le m-1} |F_j| \le \Big(\sum_{j=0}^{m-1} F_j^2\Big)^{1/2}.$$

Using Eq. (5) to replace the sum of squares gives the inequality

$$\left|\mathcal{C}^{A}_{\mathcal{S},\mathcal{T}}(\tau)\right| \leq \frac{2\sqrt{p}}{\pi\sqrt{m}}\left(\sum_{\substack{1\leq r\leq m-1\\ r\text{ odd}}}\left|L(1,\overline{\chi}^{r})\right|^{2}\right)^{1/2}.$$

Finally, applying the classical bound $L(1,\psi) \ll \log p$ for any nontrivial primitive Dirichlet character ψ modulo p [21], we obtain

$$\left|\mathcal{C}^{A}_{\mathcal{S},\mathcal{T}}(\tau)\right| \ll \frac{2\sqrt{p}}{\pi\sqrt{m}}\cdot\sqrt{\frac{m}{2}}\log p \ll \sqrt{p}\log p,$$

this yields an upper bound for the arithmetic correlation. $\square$

This bound can be found in Theorem 1 of [15], where the upper bound is derived by converting the arithmetic correlation into exponential sums and applying the orthogonality of complete exponential sums together with classical bounds for incomplete exponential sums.

Theorem 2 states that the total squared magnitude of the parity difference vector is completely determined by the odd Fourier modes. This fact also becomes especially powerful when we assume that all F_j are small. Let us focus on the case $|F_j| \leq 3$.

Theorem 4 *Let $\mathcal{S} = (s_i)_{i\geq 0}$ and $\mathcal{T} = (t_i)_{i\geq 0}$ be two binary periodic FCSR sequences with the same odd prime connection integer p and $\mathrm{ord}_p(2)$ is odd. Then for any nontrivial arithmetic correlation, we have*

(i) $\sum_{\substack{1\leq r\leq m-1\\ r\text{ odd}}}|H_r|^2 = \frac{m^2}{4}$ *if and only if the arithmetic correlation takes only the values ± 1.*

(ii) *If $\sum_{\substack{1\leq r\leq m-1\\ r\text{ odd}}}|H_r|^2 < \frac{m^2}{4} + 12m$, then the values of the arithmetic correlation belong to the set $\{\pm 1, \pm 3\}$.*

Proof Let $\Delta = \sum_{j=0}^{\frac{m}{2}-1}\frac{F_j^2-1}{8}$, by Lemma 1 (i), since F_j is odd, $F_j^2 \equiv 1 \pmod 8$, and hence Δ is a nonnegative integer. By Lemma 3, we have

$$\sum_{j=0}^{m-1}F_j^2 = 2\sum_{j=0}^{\frac{m}{2}-1}F_j^2 = 2(\frac{m}{2}+8\Delta) = m + 16\Delta.$$

Then by Theorem 2, $\sum_{\substack{1\leq r\leq m-1\\ r\text{ odd}}}|H_r|^2 = \frac{m}{4}(m+16\Delta) = \frac{m^2}{4} + 4m\Delta$.

(i) Since $\Delta \geq 0$, the smallest possible value of $\sum_{\substack{1\leq r\leq m-1\\ r\text{ odd}}}|H_r|^2$ is $\frac{m^2}{4}$, which is achieved if and only if all $F_j = \pm 1$. The conclusion then follows from Corollary 5 (i).

(ii) Suppose there exists some j_0 such that $\left|F_{j_0}\right| \geq 5$. Then $F_{j_0}^2 \geq 25$, and thus $\Delta \geq 3$, which implies $\sum_{\substack{1\leq r\leq m-1\\ r\text{ odd}}}|H_r|^2 \geq \frac{m^2}{4} + 12m$, a contradiction. The conclusion then follows from Corollary 5 (ii). $\square$

Theorem 4 transforms the problem of describing small arithmetic correlation values into a problem concerning constraints on the H_r. For example, for the connection integer $p = 7$, we have $d = 3$, $m = 2$, and the coset parity differences are $F_0 = 1$, $F_1 = -1$. Hence $F_j = 1$ or -1 for all j, and $|H_1|^2 = 1 = \frac{m^2}{4}$, consistent with Theorem 4 (i). Arithmetic correlation values are only taken ± 1.

4 The Case of d Odd and $m \in \{2, 4, 6\}$

The structural theory developed in Section 3 describes the possible distributions of parity differences when d is odd. In the subsequent, we analysis the case $m = 2, 4, 6$. The arithmetic correlation properties of binary sequences can be related to the class numbers of imaginary quadratic fields. In particular, for certain sequences, the parameter can be expressed in terms of the class number of a field of the form $\mathbb{Q}(\sqrt{-D})$ with $D > 0$. The study of class numbers of quadratic fields is a classical topic in algebraic number theory, dating back to the work of Gauss.

Let D be a positive square-free integer. Since the polynomial $t^2 + D$ is irreducible over the rationals $\mathbb{Q}$, adjoining one of its roots $\sqrt{-D}$ to $\mathbb{Q}$ gives the field extension

$$\mathbb{Q}(\sqrt{-D}) = \{a + b\sqrt{-D} \mid a, b \in \mathbb{Q}\},$$

which is called an imaginary quadratic field (see [22] for details). The order of the ideal class group of $\mathbb{Q}(\sqrt{-D})$ is called its class number and is denoted by $h(-D)$.

p is an odd prime, let $d = \mathrm{ord}_p(2)$ be odd and $m \equiv 2 \pmod 4$. Since the order of χ is m, it follows that $\chi^{\frac{m}{2}}$ is a quadratic character modulo p. On one hand, because $2 \in D$, we have $\chi^{\frac{m}{2}}(2) = 1$, meaning that 2 is a quadratic residue modulo p, and thus $p \equiv \pm 1 \pmod 8$. On the other hand, from $d - 1 = md$, it follows that $p \equiv 3 \pmod 4$. Hence, we have $p \equiv 7 \pmod 8$.

Proposition 5 *Let $d = \mathrm{ord}_p(2)$ be an odd number and $m \equiv 2 \pmod 4$, then we have $\sum_{j=0}^{\frac{m}{2}-1} (-1)^j F_j = h(-p)$.*

Proof Let $r_0 = \frac{m}{2}$. Since $m \equiv 2 \pmod 4$, r_0 is an odd number. Compute H_{r_0}, according to Eq. (3) and Lemma 3, we have

$$\begin{aligned} H_{r_0} &= \frac{1}{2} \sum_{j=0}^{m-1} F_j (-1)^j \\ &= \frac{1}{2} \sum_{j=0}^{\frac{m}{2}-1} \left[F_j (-1)^j + F_{j+\frac{m}{2}} (-1)^{j+\frac{m}{2}} \right] \\ &= \sum_{j=0}^{\frac{m}{2}-1} (-1)^j F_j. \end{aligned}$$

On the other hand, From Eq. (2), we have $H_{r_0} = \sum_{n=1}^{(p-1)/2} \chi^{r_0}(n)$. Since χ^{r_0} is a quadratic character modulo p and $p \equiv 7 \pmod 8$, it follows from Theorem 4 in Section 5.4 of [22] that $h(-p) = H_{r_0}$. This completes the proof. $\square$

Proposition 5 provides a constraint between F_j and $h(-p)$, which can be used to rule out certain prime connection integers without small arithmetic correlation values.

Corollary 6 Let $d = \mathrm{ord}_p(2)$ be an odd number and $m \equiv 2 \pmod 4$. Let B is an odd positive integer, if $|F_j| \le B$ holds for all j, then $h(-p) \le \frac{mB}{2}$.

Proof By Proposition 5, we have

$$h(-p) = \left| \sum_{j=0}^{\frac{m}{2}-1} (-1)^j F_j \right| \le \sum_{j=0}^{\frac{m}{2}-1} \left| F_j \right|,$$

if all $\left| F_j \right| \le B$, then $h(-p) \le \frac{mB}{2}$. $\square$

4.1 The case d odd and $m = 2$

When $m = 2$, this is the well-known half-ℓ-sequence, for which related studies can be found in [12–14].

Proposition 6 *All nontrivial arithmetic correlation values of any binary half-ℓ-sequence with connection integer p belong to the set $\{h(-p), -h(-p)\}$.*

Proof According to Proposition 5, we have $F_0 = h(-p)$. It then follows from Lemma 3 that $F_1 = -F_0 = -h(-p)$. The proof is completed by Proposition 1. $\square$

According to [23], we know that $h(-D) = 1$ only for $D = 3, 4, 7, 8, 11, 19, 43, 67, 163$. From the table of class numbers of imaginary quadratic fields in [22], for $p < 500$, we know that the primes p such that $h(-p) = 3$ are $23, 31, 59, 83, 107, 139, 211, 283, 307, 331, 379, 499$. Hence we have the following conclusions.

Corollary 7 Let $\mathcal{S}$ and $\mathcal{T}$ be two binary half-ℓ-sequences with prime connection integer p and $p \equiv 7 \pmod 8$. When $\mathcal{S}$ and $\mathcal{T}^{(\tau)}$ are distinct,
(i) $\mathcal{C}^A_{\mathcal{S},\mathcal{T}}(\tau) \in \{\pm 1\}$ holds only if $p = 7$;
(ii) If $p = 23$, then $\mathcal{C}^A_{\mathcal{S},\mathcal{T}}(\tau) \in \{\pm 3\}$.

4.2 Nonexistence for odd d and $m = 4$

We consider the next even $m = 4$. However, for FCSR sequences with prime connection integers, d odd and $m = 4$ turns out to be impossible.

Proposition 7 *There exists no odd prime p such that $d = \mathrm{ord}_p(2)$ is odd and $m = \frac{p-1}{d} = 4$.*

Proof Assume such a prime p exists, then $\frac{p-1}{d} = 4$, since d is odd, we get $p \equiv 5 \pmod 8$. Write $p = 8k + 5$, we have $p^2 = 64k^2 + 80k + 25$, then $\frac{p^2-1}{8} = 8k^2 + 10k + 3$ is odd. Hence $\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8} = -1$, that is 2 is a quadratic nonresidue modulo p. By Euler's criterion, $2^{(p-1)/2} \equiv -1 \pmod p$. Since $p-1 = 4d$, we have $2^{2d} \equiv -1 \pmod p$. A contradiction follows from $2^d \equiv 1 \pmod p$. $\square$

Corollary 8 Let p be an odd prime with $m = (p-1)/\operatorname{ord}_p(2) = 4$. Then $\operatorname{ord}_p(2)$ is even, and any two binary FCSR sequences $\mathcal{S}$ and $\mathcal{T}$ with the same connection integer p have ideal arithmetic correlation.

Proof See Theorem 3 in [15]. □

Since d is odd and $m = 4$ is impossible, the next admissible even index greater than 2 is $m = 6$. In what follows, we present a Fourier analysis for $m = 6$ and illustrate it with the connection prime $p = 31$, which provides a nontrivial example of a binary FCSR sequence whose arithmetic correlation values are all small.

4.3 The case d odd and $m = 6$

Assume p is an odd prime with $d = \operatorname{ord}_p(2)$ odd and $m = \frac{p-1}{d} = 6$. Let χ be a multiplicative character of order 6 with kernel $D = \langle 2 \rangle$. Let $\zeta_6 = e^{\frac{2\pi i}{6}}$ and write $F_0 = a$, $F_1 = b$, $F_2 = c$. Then by Lemma 3, $(F_0, F_1, F_2, F_3, F_4, F_5) = (a, b, c, -a, -b, -c)$.

By Eq. (3), for the odd indices $r \in \{1, 3, 5\}$, we have

$$H_r = a + b\zeta_6^r + c\zeta_6^{2r}. \tag{6}$$

We get $H_3 = a - b + c$ and $H_5 = \overline{H_1}$.

Under the above conditions, we have $p \equiv 7 (\operatorname{mod} 24)$ and χ^3 is a quadratic character modulo p. According to Proposition 5, we have $H_3 = h(-p)$.

Theorem 8 *In the case $m = 6$, the arithmetic correlation values of the sequence with connection integer p are not all equal to 1 or -1.*

Proof Assuming $F_j \in \{1, -1\}$ for $0 \le j \le 5$, we have $a, b, c \in \{1, -1\}$. Since $h(-p) = H_3 = a - b + c$, it follows that $a - b + c \in \{-3, -1, 1, 3\}$. Since $h(-p)$ is a positive integer, we can only have $h(-p) = 1$ or $h(-p) = 3$. When $h(-p) = 1$, the primes satisfying $h(-p) = 1$ can only be $3, 7, 11, 19, 43, 67, 163$. Since $p \equiv 7 (\operatorname{mod} 24)$, the only possibility is 7, but in that case $m \neq 6$, which leads to a contradiction. When $h(-p) = 3$, the only possibility is $a = 1, b = -1, c = 1$. From Eq. (6), we obtain $H_1 = 0$. From Eq. (4), $G(\chi) \neq 0$ and $L(1, \bar{\chi}) \neq 0$ [20], we obtain $H_1 \neq 0$, which leads to a contradiction. By Proposition 1, the proof is complete. □

Corollary 9 In the case $m = 6$, if the arithmetic correlation values of the sequence with connection integer p are taken from $\{1, -1, 3, -3\}$, then it must hold that $h(-p) \in \{3, 5, 7\}$.

Proof Let N_3 denote the number of indices j with $|F_j| = 3$, where $0 \le j \le 5$. By Lemma 3, N_3 is an even number. Moreover, by Theorem 8, we have $N_3 \in \{2, 4, 6\}$. Since $\sum_{j=0}^{5} F_j^2 = 6 + 8N_3$, and by Theorem 2, $H_5 = \overline{H_1}$ and $H_3 = h(-p)$, it follows that $2|H_1|^2 = 9 + 12N_3 - (h(-p))^2$. From Eq. (4), $G(\chi) \neq 0$ and $L(1, \bar{\chi}) \neq 0$ [20], we obtain $H_1 \neq 0$, which implies that $h(-p) < 9$. Since $h(-p) = a - b + c$, and $a, b, c \in \{\pm 1, \pm 3\}$, it follows that $h(-p) \in \{1, 3, 5, 7\}$.

Moreover, among the cases with $h(-p) = 1$, only $p = 7$ satisfies $p \equiv 7 \pmod{24}$. However, in this case, $m = 2$, and thus $h(-p) \in \{3, 5, 7\}$. $\square$

For example, let connection integer $p = 31$. $d = \mathrm{ord}_{31}(2) = 5$, $m = 6$. It is known that $h(-31) = 3$. The parity difference vector is

$$(F_0, F_1, F_2, F_3, F_4, F_5) = (3, 1, 1, -3, -1, -1),$$

Thus, $a - b + c = 3 = h(-31)$. This indicates that when $h(-p) = 3$, there exists a prime connection integer such that the arithmetic correlation values of the sequence belong to $\{\pm 1, \pm 3\}$.

Next, we list all possible patterns for a, b, c taken from $\{\pm 1, \pm 3\}$ under the conditions $m = 6$ and $h(-p) = 3, 5, 7$. The case $H_1 = 0$ must be excluded. From Eq. (6), it is easy to see that $H_1 = 0$ if and only if $(a, b, c) = (t, -t, t)$.

Proposition 9 *Let $m = 6$ and d be an odd number. If all the F_j belong to $\{\pm 1, \pm 3\}$, then $h(-p) \in \{3, 5, 7\}$. When $(F_0, F_1, F_2) = (a, b, c)$, the candidate patterns for (a, b, c) are:*
(i) *When $h(-p) = 3$, the candidates are exactly $(-3, -3, 3)$; $(-1, -3, 1)$; $(1, -3, -1)$; $(3, -3, -3)$; $(-1, -1, 3)$; $(3, -1, -1)$; $(1, 1, 3)$; $(3, 1, 1)$; $(3, 3, 3)$.*
(ii) *When $h(-p) = 5$, the candidates are exactly $(-1, -3, 3)$; $(1, -3, 1)$; $(3, -3, -1)$; $(1, -1, 3)$; $(3, -1, 1)$; $(3, 1, 3)$.*
(iii) *When $h(-p) = 7$, the candidates are exactly $(1, -3, 3)$; $(3, -3, 1)$; $(3, -1, 3)$.*

Proposition 9 identifies candidate patterns based on the two conditions $a - b + c = h(-p)$ and $H_1 \neq 0$. The listed candidates are only necessary conditions, not sufficient ones, and they may not all be realizable by a specific prime p. Nevertheless, the analysis above narrows the search space for connecting integers that yield small arithmetic correlation values.

5 Conclusion

In this work, we have developed a general structural description of the arithmetic correlation of binary FCSR sequences when the connection integer p is an odd prime and $d = \mathrm{ord}_p(2)$ is odd. The following conclusions are drawn:

Oddness and boundedness: Every nonzero shift arithmetic correlation value is an odd integer whose absolute value does not exceed d.

Antisymmetry: The parity difference vector satisfies $F_{j+m/2} = -F_j$, which forces each magnitude to appear equally often with positive and negative signs.

Fourier constraints: A Parseval-type identity relates the total squared magnitude of the parity differences to the odd Fourier modes, providing an upper bound for the arithmetic correlation and explicit constraints on the distribution when the only possible values are ± 1 and ± 3.

Specifically, we analyze cases where the index of cosets is 2, 4, or 6, examining situations in which the arithmetic correlation of the sequence associated with a prime connecting integer p takes small values. Together, these results give a framework that explains the observed correlation patterns, and the theory is general for any prime p

with odd d. Future work may extend these structural results to more odd numbers, and may explore applications in sequence design and cryptanalysis where precise control of correlation is required.

Declarations

- Funding
  This work is supported by the National Natural Science Foundation of China (No. 62272420), Natural Science Foundation of Fujian Province (No. 2023J01535), Fujian Key Laboratory of Financial Information Processing (Putian University) (NO. JXJS202502).
- Conflict of interest
  The authors declare no competing interests.

References

[1] Golomb, S.W., Gong, G.: Signal Design for Good Correlation: For Wireless Communication, Cryptography, and Radar. Cambridge University Press, Cambridge, UK (2005)

[2] Golomb, S.W.: Shift Register Sequences. Aegean Park Press, Laguna Hills, CA (1967)

[3] Goresky, M., Klapper, A.: Algebraic Shift Register Sequences. Cambridge University Press, Cambridge, UK (2012)

[4] Klapper, A., Goresky, M.: Feedback shift registers, 2-adic span, and combiners with memory. Journal of Cryptology **10**(2), 111–147 (1997)

[5] Klapper, A., Xu, J.Z.: Algebraic feedback shift registers. Theoretical Computer Science **226**(1-2), 61–92 (1999)

[6] Mandelbaum, D.: Arithmetic codes with large distance. IEEE Transactions on Information Theory **13**(2), 237–242 (1967)

[7] Goresky, M., Klapper, A.: Arithmetic crosscorrelations of feedback with carry shift register sequences. IEEE Transactions on Information Theory **43**(4), 1342–1345 (1997)

[8] Chen, Z.X., Niu, Z.H., Sang, Y.Q., Wu, C.H.: Arithmetic autocorrelation of binary m-sequences. Cryptologia **47**(5), 449–458 (2023)

[9] Jing, X.Y., Zhang, A.X., Feng, K.Q.: Arithmetic autocorrelation distribution of binary m-sequences. IEEE Transactions on Information Theory **69**(9), 6040–6047 (2023)

[10] Jing, X.Y., Feng, K.Q.: Arithmetic crosscorrelation of binary m-sequences with coprime periods. Finite Fields and Their Applications **96**, 102424 (2024)

[11] Jing, X.Y., Chen, Z.X., Feng, K.Q.: On arithmetic autocorrelation of p-ary m-sequences. Advances in Mathematics of Communications **19**(5), 1518–1538 (2025)

[12] Chen, Z.X., Edemskiy, V., Niu, Z.H., Sang, Y.Q.: Arithmetic correlation of binary half-ℓ-sequences. IET Information Security **17**(2), 289–293 (2023)

[13] Xiao, L.M., Edemskiy, V.: Arithmetic Autocorrelation of Certain Binary Half-ℓ-Sequences. In: International Conference on Information Security and Cryptology. Springer, Singapore (2025)

[14] Edemskiy, V.: Notes about imbalance and arithmetic autocorrelation of N-ary half-ℓ-sequences. Cryptologia **50**(1), 81–93 (2026)

[15] Chen, Z.R., Edemskiy, V., Wu, C.H.: Binary sequences with ideal and small arithmetic correlations. Cryptography and Communications **17**(2), 453–464 (2025)

[16] Edemskiy, V.: On the ideal arithmetic autocorrelation of binary FCSR sequences. Problems of Information Transmission **61**(2), 211–218 (2025)

[17] Lidl, R., Niederreiter, H.: Finite Fields. Cambridge University Press, Cambridge, UK (1997)

[18] Stein, E.M., Shakarchi, R.: Fourier Analysis: An Introduction. Princeton University Press, Princeton, NJ (2003)

[19] Berndt, B.C.: Classical theorems on quadratic residues. Enseignement Mathématique **22**(3-4), 261–304 (1976)

[20] Ireland, K.F., Rosen, M.I.: A Classical Introduction to Modern Number Theory. Springer, New York, NY (1990)

[21] Apostol, T.M.: Introduction to Analytic Number Theory. Springer, New York, NY (1976)

[22] Borevich, Z.I., Shafarevich, I.R.: Number Theory. Academic Press, New York, NY (1966)

[23] Goldfeld, D.: Gauss' class number problem for imaginary quadratic fields. Bulletin of the American Mathematical Society **13**(1), 23–37 (1985)